

HEART DISEASE PREDICTION USING MACHINE LEARNING

INTERNSHIP PROJECT REPORT

Submitted By

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SANIKA NARALE

ABSTRACT

Heart disease is one of the leading causes of death worldwide and early diagnosis plays a crucial role in reducing mortality rates. The objective of this project is to develop a Machine Learning-based system that can predict the presence of heart disease using various medical parameters.

The dataset used in this project consists of 303 patient records with 14 attributes, including age, sex, chest pain type, cholesterol level, resting blood pressure, maximum heart rate achieved, and other health indicators. Data preprocessing techniques were applied to clean and prepare the dataset for analysis.

Various data visualization techniques such as Heart Disease Distribution, Age Distribution, Gender vs Heart Disease, Correlation Heatmap, and Confusion Matrix were used to understand patterns and relationships within the data. Logistic Regression and Random Forest algorithms were implemented and compared to evaluate their performance.

The Logistic Regression model achieved an accuracy of 86.89%, while the Random Forest model achieved an accuracy of 83.61%. Based on the comparison, Logistic Regression was selected as the final prediction model. The developed system successfully predicts whether a patient is likely to have heart disease or not.

This project demonstrates the practical application of Artificial Intelligence and Machine Learning in healthcare and highlights the potential of predictive analytics in supporting medical decision-making and early disease detection.

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INTRODUCTION

Heart disease is one of the leading causes of death worldwide and poses a significant challenge to healthcare systems. Early diagnosis and timely treatment can greatly improve patient outcomes and reduce the risk of severe complications. However, accurately identifying heart disease at an early stage can be difficult due to the involvement of multiple medical factors and symptoms.

With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), predictive models can be developed to assist healthcare professionals in diagnosing diseases more efficiently. Machine Learning algorithms are capable of analyzing large volumes of medical data, identifying hidden patterns, and making accurate predictions based on historical records.

The Heart Disease Prediction System is a Machine Learning-based project designed to predict whether a patient is likely to have heart disease using various health-related parameters. The dataset used in this project contains information such as age, sex, chest pain type, cholesterol level, resting blood pressure, maximum heart rate achieved, fasting blood sugar, and other clinical attributes.

In this project, data preprocessing and visualization techniques were applied to understand the dataset and identify important relationships between different features. Logistic Regression and Random Forest algorithms were implemented and evaluated to determine the most effective prediction model. The performance of these models was measured using accuracy scores, classification reports, confusion matrices, and graphical visualizations.

The final prediction system can classify patients into heart disease and non-heart disease categories based on their medical information. Such systems can assist healthcare professionals in making informed decisions and contribute to the early detection and prevention of cardiovascular diseases.

PROBLEM STATEMENT

Heart disease is one of the most common causes of death worldwide. Diagnosing heart disease at an early stage is essential for effective treatment and prevention. Traditional diagnostic methods often require extensive medical examinations and expert analysis, which can be time-consuming and costly.

Healthcare organizations generate large amounts of patient data containing valuable information that can be used for predictive analysis. By applying Machine Learning techniques to this data, it is possible to identify patterns and predict the likelihood of heart disease in patients.

The main problem addressed in this project is to develop a Machine Learning model that can accurately predict whether a patient is likely to have heart disease based on various medical attributes and health indicators.

OBJECTIVES

The primary objectives of this project are:

1. To analyze and understand the heart disease dataset.
2. To perform data preprocessing and data cleaning techniques.
3. To visualize the dataset using graphical representations.
4. To build Machine Learning models for heart disease prediction.
5. To compare the performance of different algorithms.
6. To evaluate model performance using accuracy metrics and confusion matrices.
7. To develop a predictive system capable of identifying heart disease risk.
8. To demonstrate the application of Artificial Intelligence and Machine Learning in healthcare.

DATASET DESCRIPTION

The dataset used in this project is a Heart Disease Dataset containing medical information of patients. The dataset consists of 303 records and 14 attributes that describe different health parameters used for predicting heart disease.

The attributes included in the dataset are:

- Age – Age of the patient
- Sex – Gender of the patient
- Chest Pain Type (cp) – Type of chest pain experienced
- Resting Blood Pressure (trestbps) – Blood pressure measured at rest
- Cholesterol (chol) – Serum cholesterol level
- Fasting Blood Sugar (fbs) – Blood sugar level after fasting
- Resting ECG (restecg) – Electrocardiographic results
- Maximum Heart Rate Achieved (thalach)
- Exercise Induced Angina (exang)
- ST Depression (oldpeak)
- Slope – Slope of peak exercise ST segment
- Number of Major Vessels (ca)
- Thal – Thalassemia status
- Target – Heart Disease Status

The target attribute is used for classification. A value of 1 indicates the presence of heart disease, while a value of 0 indicates the absence of heart disease.

METHODOLOGY

The following methodology was adopted to develop the Heart Disease Prediction System:

1. Data Collection

The heart disease dataset was collected and imported into the Jupyter Notebook environment for analysis.

2. Data Preprocessing

The dataset was examined for missing values, inconsistencies, and data quality issues. Necessary preprocessing steps were applied.

3. Exploratory Data Analysis (EDA)

Various graphs and visualizations were generated to understand the distribution of data and relationships among features.

4. Data Splitting

The dataset was divided into training and testing sets using the train-test split technique.

5. Model Building

Machine Learning algorithms such as Logistic Regression and Random Forest were implemented.

6. Model Evaluation

The performance of the models was evaluated using accuracy scores, classification reports, and confusion matrices.

7. Prediction System

A prediction system was developed to classify patients as having heart disease or not based on their medical information.

HEART DISEASE DISTRIBUTION & AGE DISTRIBUTION

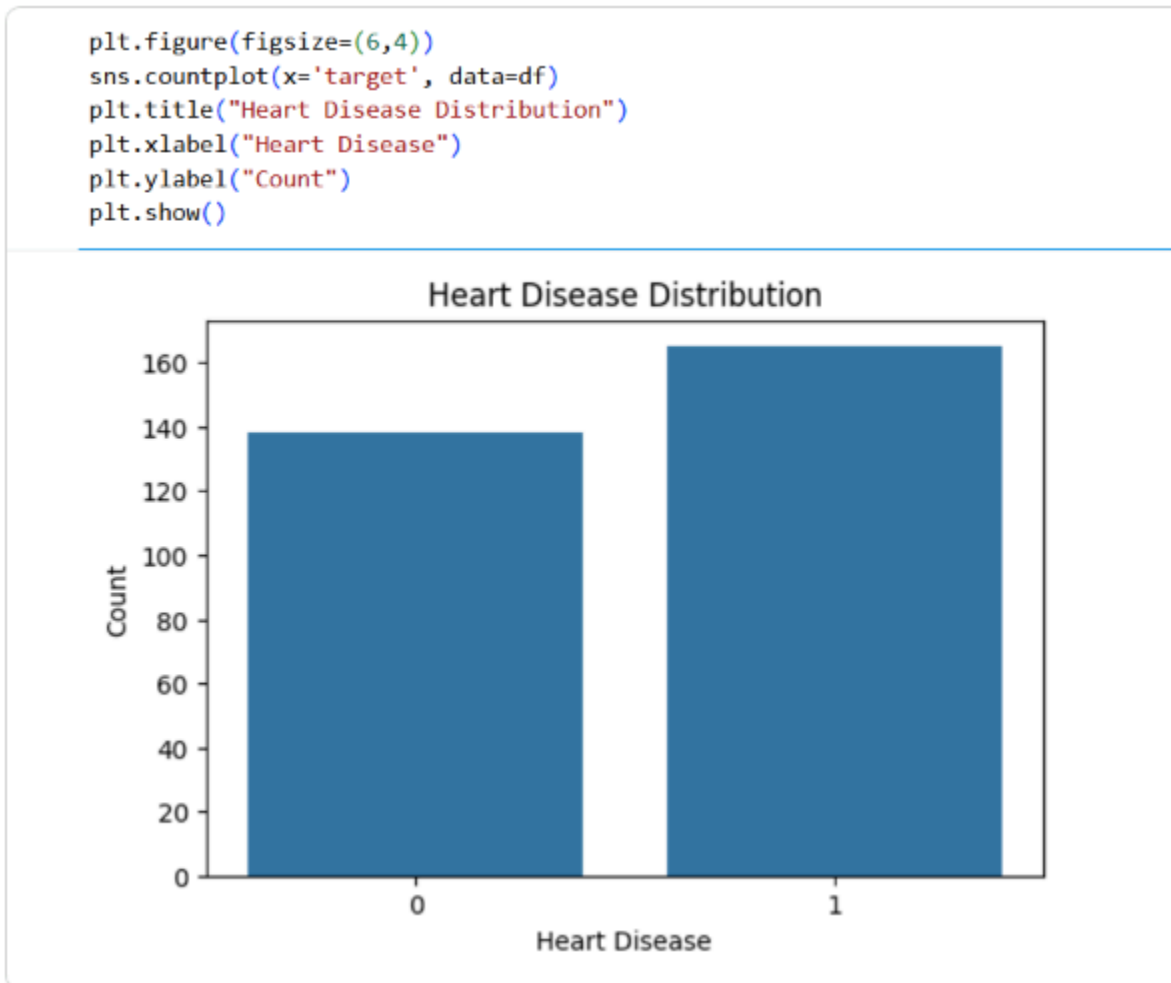


Figure 1: Heart Disease Distribution

The graph represents the number of patients with and without heart disease. It helps understand the balance of classes present in the dataset.

The value **0** represents patients who do not have heart disease, while **1** represents patients who have heart disease

```
plt.figure(figsize=(8,5))
sns.histplot(df['age'], bins=20, kde=True)

plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```

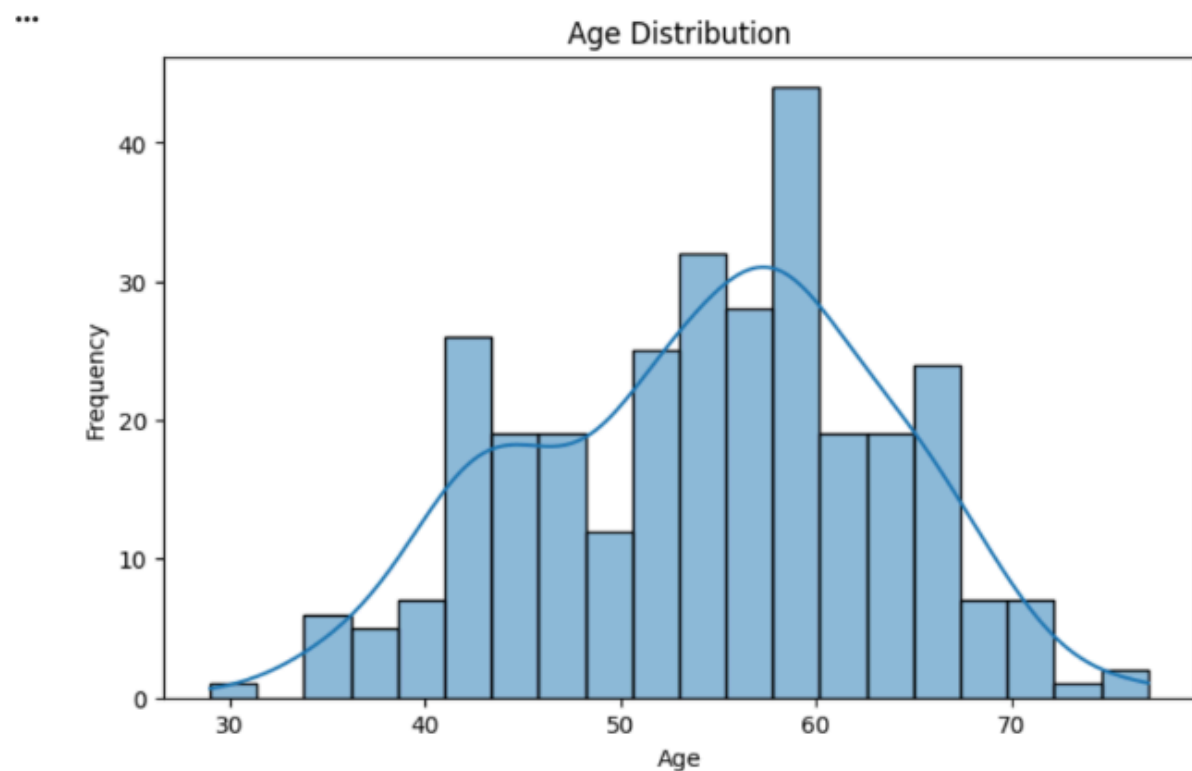


Figure 2: Age Distribution

The age distribution graph shows the spread of patient ages within the dataset. Most patients belong to the middle-age and older-age groups.

The Age Distribution graph illustrates the frequency of patients belonging to different age groups. Most of the patients are concentrated between the ages of **45 and 65 years**, indicating that heart disease-related data is more prevalent among middle-aged and older individuals.

CORRELATION HEATMAP & GENDER VS HEART DISEASE

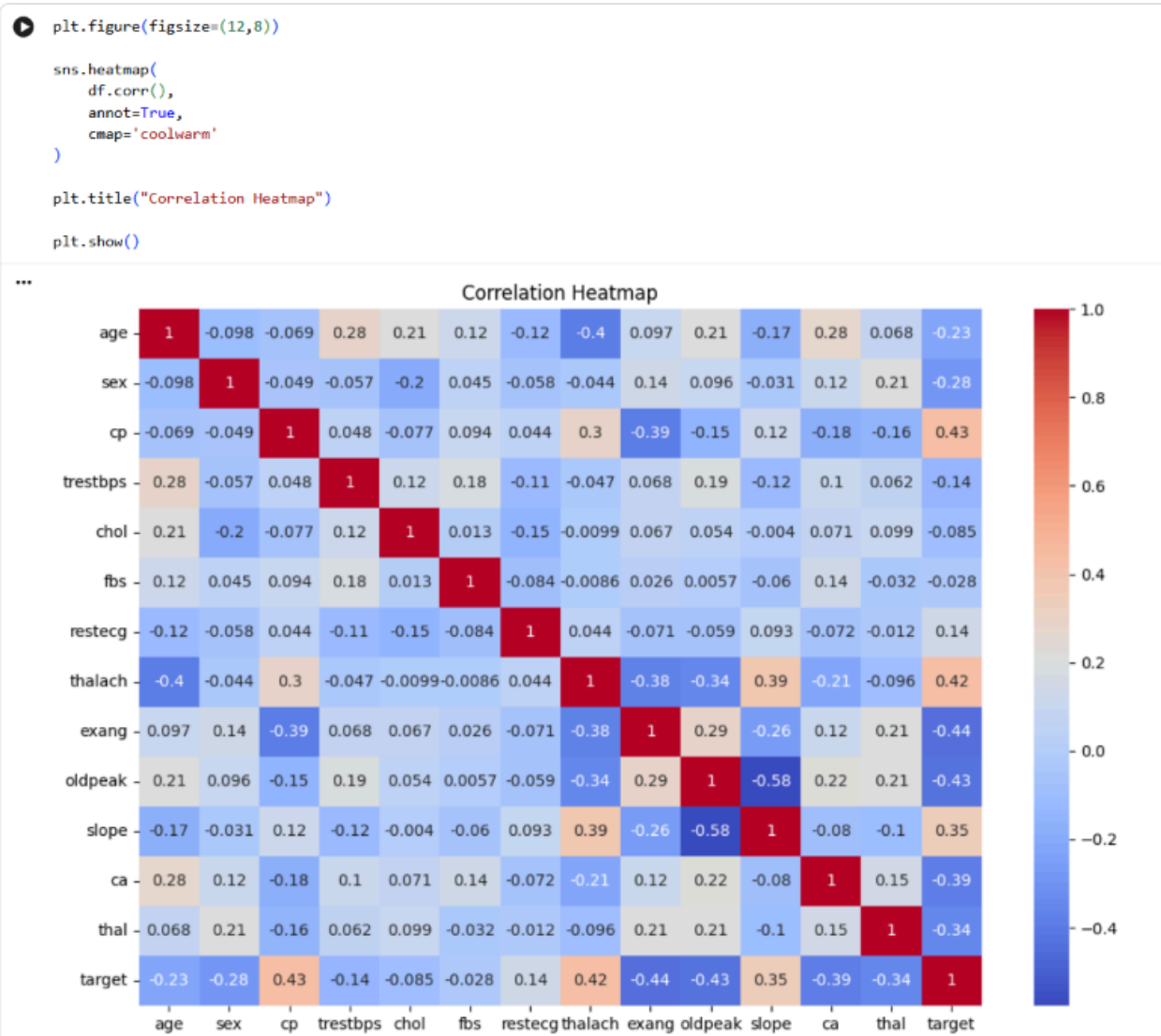


Figure 3: Correlation Heatmap

The heatmap illustrates relationships among dataset attributes. Positive and negative correlations help identify influential features.

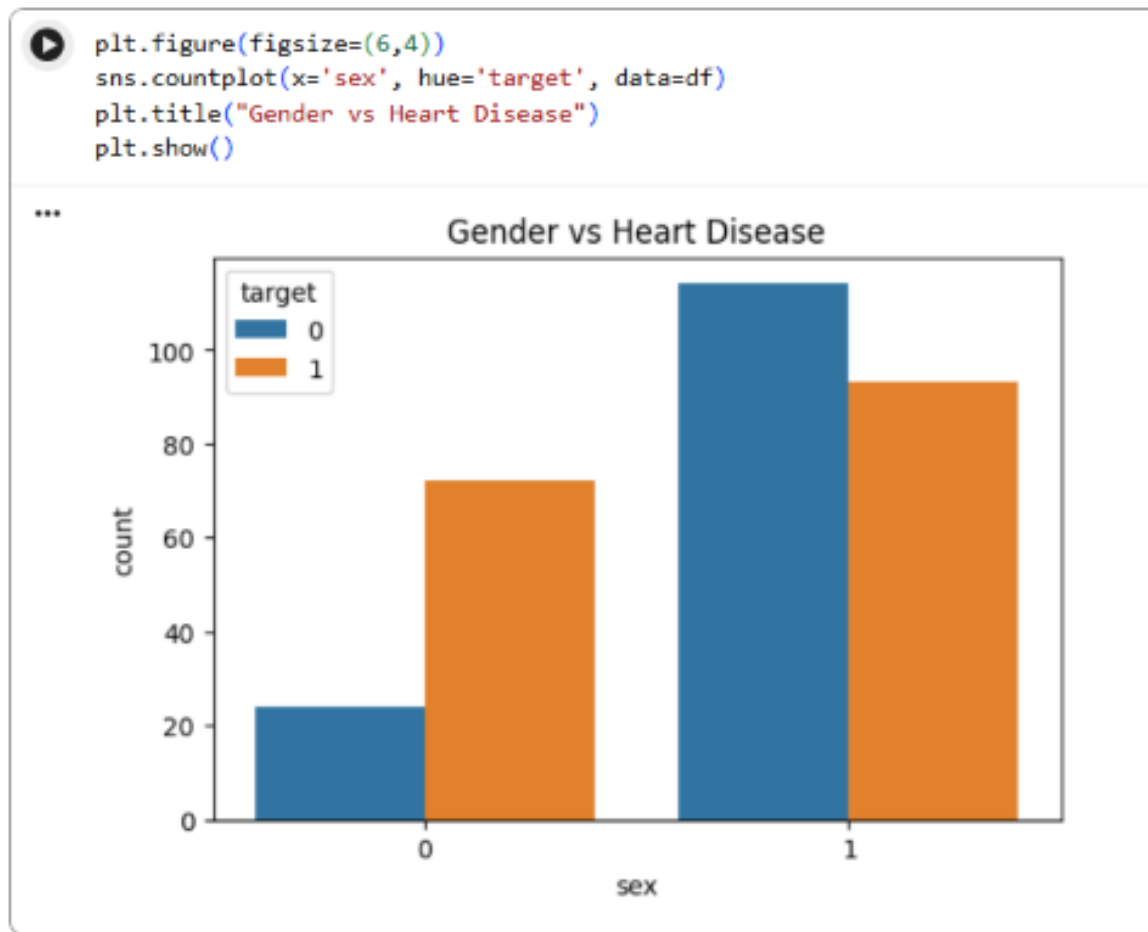


Figure 4: Gender vs Heart Disease

This graph compares heart disease occurrences among different genders and helps visualize patterns within the dataset.

MODEL BUILDING

Two Machine Learning algorithms were implemented in this project:

1. Logistic Regression

Logistic Regression is a supervised Machine Learning algorithm used for binary classification problems. It predicts whether a patient has heart disease or not based on the input features.

2. Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Both algorithms were trained using the training dataset and evaluated using the testing dataset. Their performance was compared based on accuracy and prediction capability.

CONFUSION MATRIX & MODEL COMPARISON

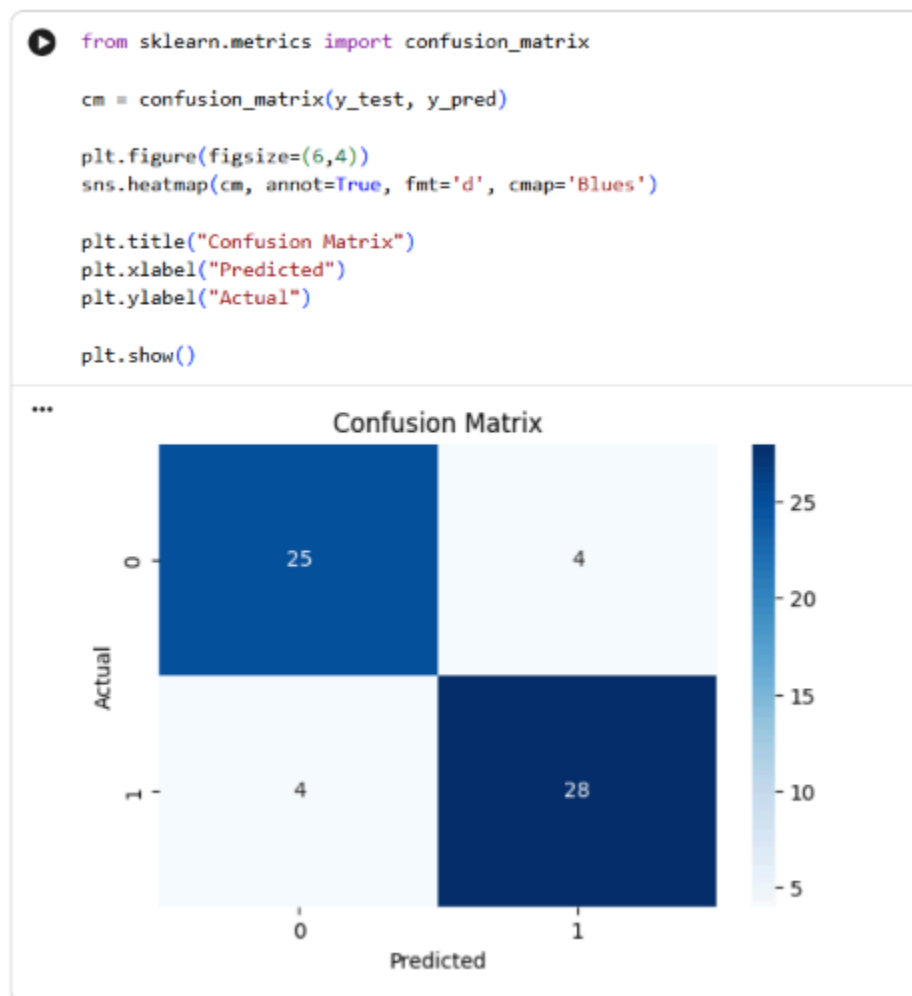


Figure 5: Confusion Matrix

The confusion matrix evaluates the classification performance of the Logistic Regression model by comparing actual and predicted values.

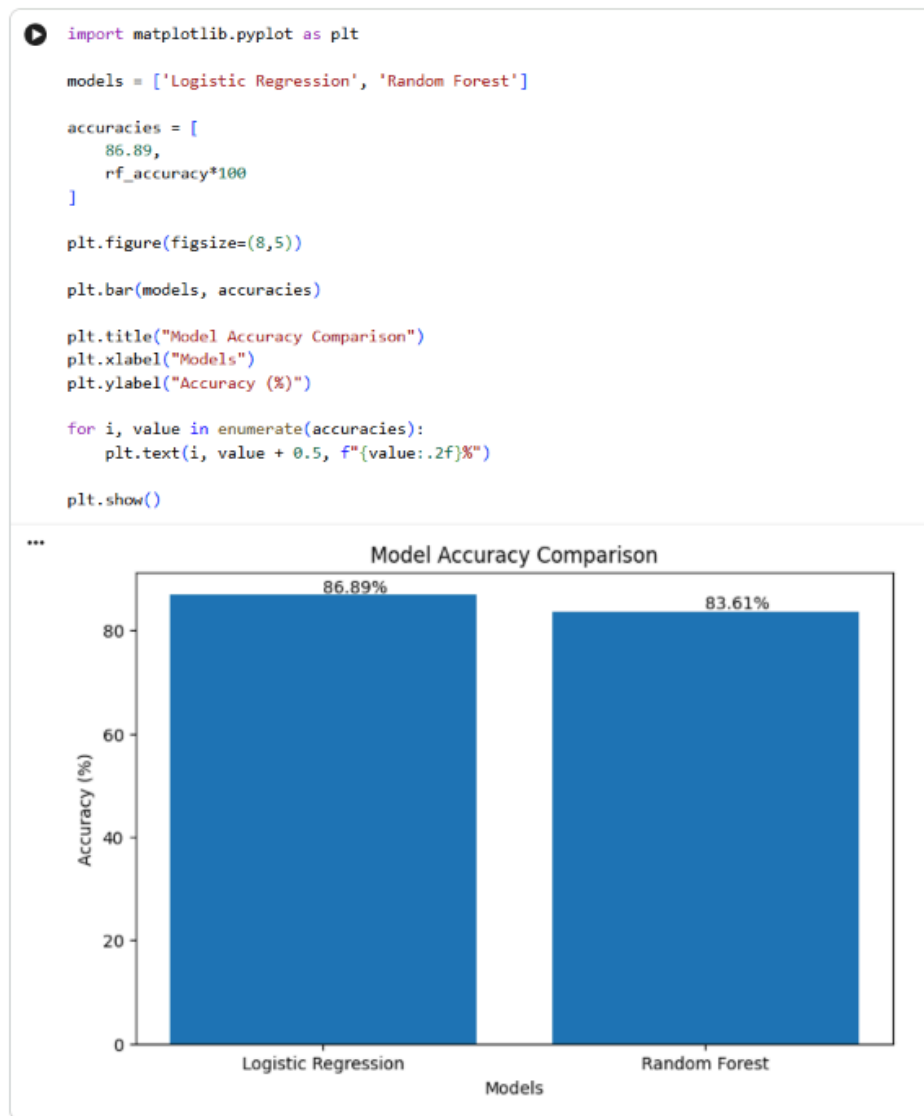


Figure 6: Model Accuracy Comparison

The graph compares the performance of Logistic Regression and Random Forest models.

Logistic Regression Accuracy = 86.89%

Random Forest Accuracy = 83.61%

RESULTS AND DISCUSSION

The developed Machine Learning models were evaluated using the testing dataset.

Results Obtained:

- Logistic Regression Accuracy: 86.89%
- Random Forest Accuracy: 83.61%

The Logistic Regression model achieved higher accuracy compared to Random Forest and therefore was selected as the final prediction model.

The classification report and confusion matrix further confirmed the effectiveness of the Logistic Regression model in predicting heart disease.

The developed prediction system successfully classifies patients based on their medical information and demonstrates the usefulness of Machine Learning in healthcare analytics.

CONCLUSION

The Heart Disease Prediction System was successfully developed using Machine Learning techniques. The project involved data preprocessing, visualization, model training, performance evaluation, and disease prediction.

Among the implemented algorithms, Logistic Regression achieved the highest accuracy of 86.89% and was selected as the final model. The prediction system can effectively identify patients who are at risk of heart disease based on medical attributes.

This project demonstrates the practical application of Artificial Intelligence in healthcare and highlights the importance of predictive analytics in assisting medical decision-making.

FUTURE SCOPE

- Integration with hospital management systems.
- Real-time patient health monitoring.
- Mobile-based heart disease prediction applications.
- AI-powered healthcare assistance systems.
- Improved prediction accuracy using advanced Machine Learning algorithms.

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